

Waleed Rashid

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EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Engineering, Computer Engineering

Sep. 2023 – Apr. 2027

- **Relevant Coursework:** Electric Networks, Digital Systems, Electronic Circuits I, Signals & Systems II, Microprocessor Systems, Control Systems, Communication Systems, Data Structures & Algorithms, Operating Systems.

EXPERIENCE

Electrical Design Intern

Jul. 2025 – Aug. 2025

EPCM

Oakville, ON

- Updated electrical schematics based on redlines using **AutoCAD** and **EPLAN**, supporting engineering projects to ensure accurate and compliant drawings.
- Developed and maintained comprehensive electrical Bill of Materials (BOMs), improving component tracking, procurement accuracy, and overall project readiness.
- Collaborated closely with and worked under the supervision of a Professional Engineer (P.Eng.), applying industry best practices while gaining hands-on experience in real-world electrical design, documentation, and review workflows.

LEADERSHIP

Electrical Design Team Member

Sep. 2024 – Dec. 2025

Toronto MET Baja Racing

Toronto, ON

- Optimized the vehicle's electrical architecture by integrating a busbar-based power management solution, reducing wiring complexity, and improving reliability across all connected subsystems.
- Implemented a transistor-based emergency shutdown circuit to immediately disconnect power during fault or emergency conditions, significantly enhancing electrical safety and system protection.
- Gained hands-on experience soldering, wiring, and assembling electrical components for the race car, ensuring reliable, low-resistance connections in high-vibration automotive environments.

PROJECTS

HifzCompanion | *React, TypeScript, Node.js, Express, PostgreSQL, C++, ESP32*

- Building a full-stack "Islamic study progress tracker" web application to help users plan, monitor, and improve daily memorization and revision consistency in a structured learning system.
- Designing and implementing a React + TypeScript dashboard with RESTful APIs, PostgreSQL database integration, and secure JWT-based authentication for user sessions and access control.
- Currently extending the system into an IoT-enabled platform by integrating an ESP32 embedded device for real-time session tracking, progress syncing, and interactive feedback.

Mini Multithreaded HTTP Server | *C++, Linux, TCP Sockets*

- Built a multithreaded HTTP/1.1 server in C++ by implementing TCP sockets and a custom request parser, enabling concurrent handling of client connections and serving static web content on localhost.
- Improved concurrency by designing a thread pool with a synchronized task queue, reducing request-handling overhead and allowing scalable parallel processing of HTTP requests.
- Implemented static file serving by integrating filesystem I/O and MIME-type handling, ensuring correct delivery of HTML, CSS, and images with proper HTTP responses.

ALU & FSM Central Processing Unit Design | *VHDL, FPGA*

- Designed and implemented a custom 8-bit CPU on an Altera FPGA using **VHDL**, integrating an ALU, control unit, and registers to support arithmetic and logical operations with FSM-based instruction control.
- Verified system functionality by simulating CPU operations and interfacing a 7-segment display to visualize signed binary outputs, arithmetic results, and logic states for clear system validation.

State-Driven Bookstore Application | *Java, JavaFX, State Design Pattern*

- Designed and implemented a full-featured JavaFX Bookstore application using the State Design Pattern to dynamically manage customer status transitions (Silver/Gold) and encapsulate behavior based on reward points.
- Modeled system architecture using UML Use Case and Class Diagrams, translating object-oriented design principles into a scalable NetBeans implementation with clean separation of concerns.
- Built a single-window GUI application with persistent file storage (books.txt, customers.txt), implementing transactional logic for purchases, point accumulation/redemption, and real-time state updates.

TECHNICAL SKILLS

Languages/Libraries: Java, HTML, CSS, C, C++, VHDL, MATLAB, JavaScript, TypeScript, Python, SQL.

Frameworks: React, Node.js, Express.

Technologies/Tools: Eplan, Assembly, AutoCAD, Quartus, Arduino, Microsoft Office, AWS, MySQL, VS Code, Git.